

AVANT®

Operator's Manual for Attachment



Lawn Mower 1000

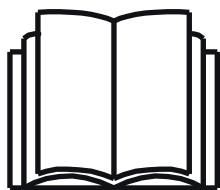
Product number

Avant 216 and 218 loaders:

A21076

Avant 220 loaders:

A21109



Please read this manual carefully before using the equipment,
and follow all instructions.

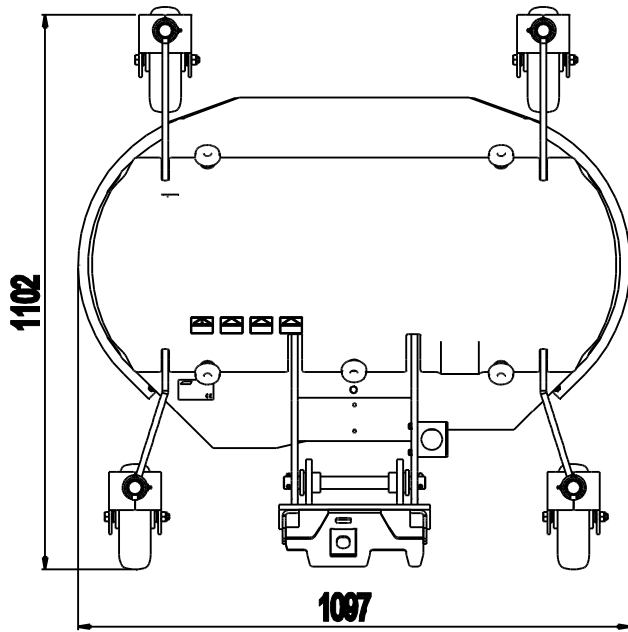
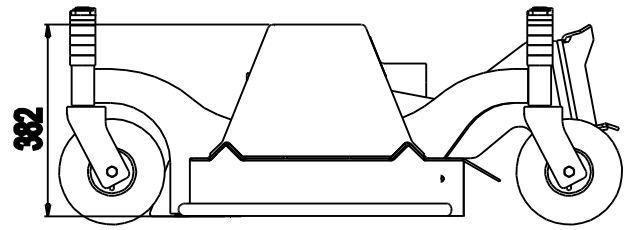
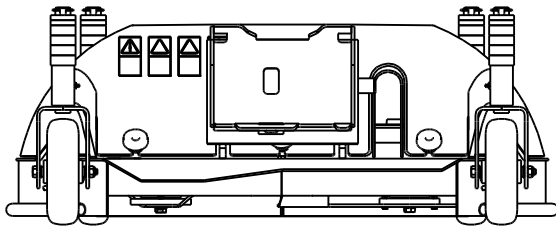
Keep this manual for later reference.

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I. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for your Avant loader. It has been designed and manufactured on the basis of years of experience on product development and manufacturing. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

The following warning symbols are used throughout this manual to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:v

	WARNING:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.

NOTICE

This signal word indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader, each attachment that are used, and all safety instructions before using the equipment. Using the equipment for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from your Avant dealer or from the manufacturer. Due to continuous product development some of the details shown in this manual may differ from your equipment. This manual contains the original instructions in English.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws, and other regulations concerning the use of the equipment. Particularly the regulations concerning the use of the equipment on public road areas must be observed. We reserve the right to change the contents of the manual without notification.

2. Designed purpose of use

The AVANT Lawn Mower 1000 is an attachment suitable for use with AVANT multi purpose loaders shown in Table 1. The lawn mower is suitable for residential and professional use as well. It is intended to be used to mow lawns on level, tended areas free of rocks, tree stumps, large branches, or other similar obstacles. The robust lawn mower can also shred tree leaves and small branches during lawn mowing. The lawn mower is equipped with mechanical floating linkage system, which allows the mower to follow the ground freely, eliminating the need for the loader boom floating.

The cutting deck has 2 blades. As standard, the mower is equipped with a mulching kit, and the clippings will be shred under the mower. The mower can be transformed to rear ejecting type, which makes it easy to cut longer grass, or allows higher mowing speed. Separate mower models are available for Avant 216/218 and 220 loaders. The two different mower models have different hydraulic components for optimal performance, and they should only be used with the loader model that they are intended for.

The lawn mower is not intended for mowing long hay shrub or equivalent taller vegetation. The lawn mower cannot be used in a hung position; it is meant to run on its own wheels<MowerType>. For mowing long hay, we recommend the AVANT Flail Mowers.

The attachment has been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks. All repair work can't be performed by the operator, and demanding repair and maintenance operations are to be left for professional maintenance. All maintenance work must be done using proper safety equipment. Spare parts must be identical with original specifications, which can be ensured by using only original spare parts. A separate spare parts catalog may be available, consult your Avant dealer.

Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment, or if you require spare parts or maintenance services.

Table 1 - Lawn mower 1000 - Compatibility with Avant loaders

Avant	216 218	220	313S 320S 320S+	419 420	520 R20 525LPG R28 528 630	R35 635 640 745 750 760
A20176	•	-	-	-	-	-
A21109	-	•	-	-	-	-

Normally the width of the mower should be equal to or greater than the width of the loader. For compatibility with a model not shown in table, contact your Avant retailer.

3. Safety instructions for the Lawn Mower

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, all operators must carefully familiarise themselves about correct use and the operator's manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse, careless use, or using an attachment that is in poor condition, may cause risk of serious injuries. Familiarise yourself with the controls of the loader, correct coupling procedure, and the correct way to operate the attachment at a safe area. Study especially how to stop the equipment in a safe manner. Read all safety precautions carefully.



Read all safety instructions carefully before handling the attachment

- When attaching an attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the attachment to the loader.** Never lift or move an unlocked attachment.
- The Lawn Mower 1000 is designed to be used by one operator at a time. Do not let others near the danger area of the equipment when it is in use.
- Never leave the driver's seat when the mower is running or if the auxiliary hydraulics control of the loader is locked on. Always keep a **safety distance of at least two metres** between persons and a mower that may start or move. Stop the mower following the safe stopping procedure and lower the mower firmly on the ground before leaving the driver's seat.
- Run the mower only when it is supported by all of its four wheels. Running the mower in any other position may expose the blades, making contact with them possible, and will also cause a hazard of flying debris.
- Do not leave the driver's seat when the loader boom is lifted. Going under a raised attachment or loader boom is dangerous. Always remember that the boom may lower due to loss of stability, mechanical fault, or if another person operates the controls of the loader. Follow safe stopping procedure before leaving the driver's seat.
- Be sure not to tilt the lawn mower too far up, since this will make it tip over towards the loader boom due to the movement made possible by the floating system linkage.
- Secure the mower against unintended movements when unlocking it from the loader. Unlock only on a level surface, and use blocks of wood to prevent the mower from moving, for example.
- Make sure that the mower is properly supported before performing any inspections, maintenance, or adjustments. If access to blades is necessary, secure the mower with suitable lifting equipment, or lower it on firm blocks of wood, for example.
- Always transport the mower as low and as close to the machine as possible to keep the centre of gravity low. On slippery surfaces drive slowly and deliberately. Operate the controls of the loader slowly and calmly especially on inclined terrain.



- Never use the mower if all covers, blade guards, or other safety related parts are not intact and securely in place. Either the side discharge guard or mulching kit must always be installed.
- Check the blades and clean the mowing deck regularly. Read the instructions about maintenance and service and use only original blades.
- Shut down the loader and place the attachment to a safe position as shown in Safe stopping procedure before any cleaning, maintenance, or adjustments.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain and other hazards, such as branches and trees that can reach to the driver's area, loose rocks, and slippery surfaces.
- Make sure to use only an attachment that is in good condition. Do not modify the attachment in any way that could affect its safety.
- Use the attachment only for its intended purpose. Other use may create unnecessary safety risks, and the equipment may get damaged.
- Make sure that the loader is equipped with necessary safety components, and that they are in working condition. Seat belt must be used. If there are specific hazards related to the operating area, use appropriate safety equipment.
- Also read the safety instructions and correct use of the loader from the operator's manual of the loader.



Improper or careless use of the attachment may cause serious risks. Familiarise yourself with the controls of the loader in a safe area. Pay particular attention to the safe stopping of the attachment and the loader; follow the safe stopping procedure shown in this manual.



Beware of the sharp blades and risk of crushing especially during maintenance or when removing blockages. Always stop the attachment by following the safe stopping procedure. Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or any component of the attachment. Keep away from the moving linkage.



Always keep a safety distance of at least two metres to other persons. Note that the mower may throw stones, gravel, or other objects found in the area at high speed. Do not allow others to get near a running mower.

Remember to wear proper protective clothing:

The noise level at the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



Wear protective gloves.



Wear safety boots whenever working with the loader.



Wear safety glasses when handling hydraulic components.

Safe stopping of the attachment, before going near the attachment:

Always stop the attachment following safe stopping procedure before leaving the driver's seat. Safe stopping procedure prevents all unintentional movements of the attachment. Note that the loader boom can move down even if the engine of the loader is turned off. Safe stopping procedure:

- Lower the boom and the attachment on the ground.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system; move all hydraulic control levers to their extreme positions a couple of times.
- Prevent starting of the machine, remove ignition key.

4. Technical specifications

Table 2 – Lawn mower 1000- Technical specifications

Product numbers	A21076	A21109
Working width:	1 000 mm	
Weight:	95 kg	
Cutting height:	30-100 mm	
Drive system:	Hydraulic, direct/belt drive	
Blades:	2 pcs, 530 mm, A33102	
Clippings handling:	Mulching (standard) Rear discharge (standard)	
Support wheel air pressure:	1,5 – 2 bar	
Recommended hydraulic flow:	20-22 l/min	24-26 l/min
Maximum input of hydraulic energy:	18,5 MPa (185 bar), 25 l/min	18,5 MPa (185 bar), 30 l/min
Compatible AVANT loaders:	See Table 1	
	216, 218	220

NOTICE

Choose a mower that is compatible with the loader. Do not exceed the maximum input of hydraulic energy or the operating pressure.

4.1 Safety labels and main components of the attachment

Listed below are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or if they have detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.



A46771



A46772



A46798



A46802

Table 3 – Decal locations and warning messages

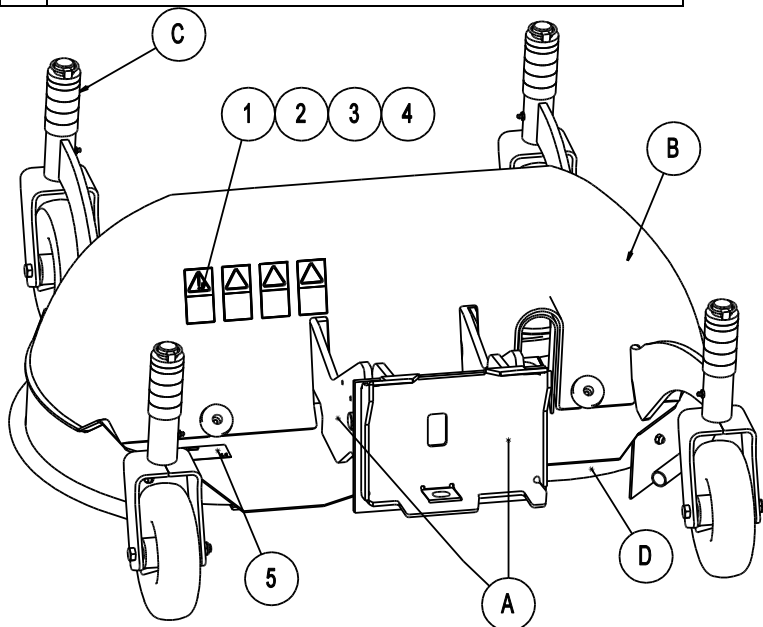
	Decal	Explanation
1	A46771	Read instructions before use.
2	A46772	Crushing hazard - Do not go under a raised attachment; stay away from raised equipment.
3	A46798	Risk of getting drawn into and crushed by belt drive components; never operate without all covers in place.
4	A46802	Beware of sharp blades – never reach to the rotating components and do not use the equipment without all covers in place.

 Avant Tecno Oy Ylötie 1 FIN-33470 YLÖJÄRVI	
TYPE	No.
	kg l/min
Max	l/min, MPa (bar)
2014	Made in EU www.avanttecno.com 

Attachment identification plate

Table 4 – Main components of lawn mower

A	Quick attach plate and floating linkage
B	Protective cover
C	Cutting height adjustment
D	Rear discharge cover



5. Connecting the Lawn Mower to the loader

Attaching the attachment to the loader is quick and easy, but it must be done carefully. The attachment is mounted to the loader boom by using the quick attach plate on the loader boom and the counterpart on the attachment.

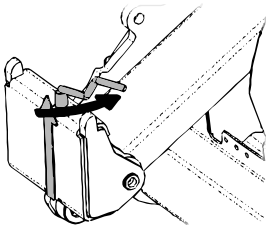
If the attachment is not locked to the loader, it may detach from the loader and cause a hazardous situation. The loader must not be driven and the boom must never be lifted when the attachment has not been locked. To prevent hazardous situations, always follow the coupling procedure shown below. Also remember the safety instructions shown in Chapter 3. The attachment is mounted to the loader as follows:



Make sure that an unlocked attachment will not move or fall over. Do not stay in the area between the attachment and the loader. Mount the attachment only on level surface.

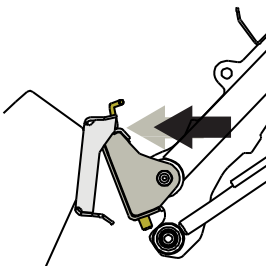
Never move or lift an attachment that has not been locked.

200 series loaders:



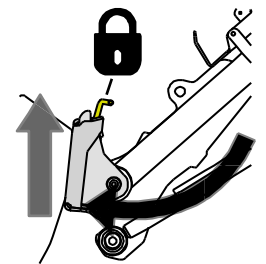
Step 1:

- Lift the quick attach plate locking pin up and turn it backwards into the slot so that it is locked the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment.
- Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.

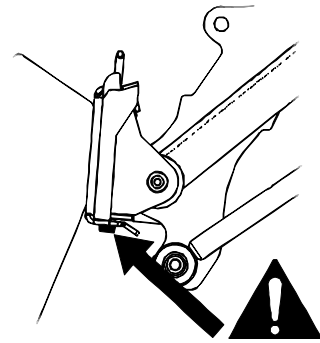


Step 3:

- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Lock the locking pin manually.
- **Always check the locking of the locking pin.**



An attachment that has not been completely locked to the loader may fall on the boom or towards the operator, or fall under the loader during driving, causing loss of control of the loader. Never move or lift an attachment that has not been locked. Before moving or lifting the attachment, make sure that the locking pin is in the lower position and comes slightly through the hole below the quick coupling plate.



Excessive tilting or lifting of an unlocked attachment increases the risk of tipping the attachment over. Do not use the automatic locking of the locking pins when the attachment is lifted more than one metre from the ground. If the locking pins do not return to the normal position when tilting, do not tilt or raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

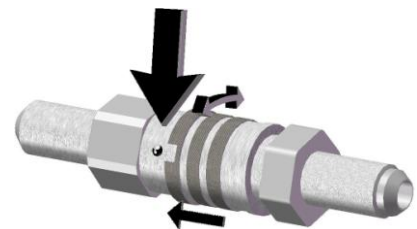
5.1 Hydrauliletkujen kiinnittäminen ja irrottaminen

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.

Hydraulic couplings in 200 series loaders:

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200 series hydraulic quick couplings, note the locking system of the fittings. There is a small groove in the collar of all female fittings, and a small ball near the collar. When connecting or disconnecting, the groove and the ball must be aligned.

The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The collar may also turn accidentally to the locking position, making it impossible to connect or disconnect the couplings.



Never connect or disconnect quick couplings or other hydraulic components while the control lever of the auxiliary hydraulics control lever is locked on or if the system is pressurized. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns. Follow safe stopping procedure before disconnecting hydraulics.

NOTICE

Keep all fittings as clean as possible; use the protective caps on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place the couplings onto the holder on the attachment.

Disconnecting hydraulic hoses:

Before disconnecting the fittings, lower the attachment to safe position on solid and level surface. Turn the control lever of the auxiliary hydraulics to its neutral position.

NOTICE

When uncoupling the attachment, always disconnect the hydraulic couplings before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader back and forth before disconnecting the couplings.

5.2 Storage

Uncouple and store the attachment on level ground, but avoid leaving the attachment laying directly on against the ground. Place it on blocks of wood or on a pallet, for example. The best way to store the attachment is a place where it is protected from direct sunlight, rain, and extreme temperatures. Make sure that the attachment is secured against all movements during storage, and never climb on the attachment.



Make sure the attachment is secured against any movement during storage. If necessary, use, for example, blocks of wood under the supports to ensure adequate stability.

Before winter storage, carefully clean the cutting deck and the mower components, removing all remaining clippings. Touch up the paint where needed and grease the lubrication points to prevent rust damage.

6. Instructions for use

Check the attachment and the operating environment once more before starting to work, and that all obstacles have been removed from the operating area. Quick inspection of the equipment and the operating area before use are parts of ensuring safety and the best performance of the equipment.

6.1 Checks before use



Before using the mower, check that it is intact and that all protective covers are in place.

NOTICE

Before each use, check the operation of the mower and ensure that the blades can rotate freely. Check that there is no string or other foreign material coiled around the blades, and that the floating linkage can move as intended.

- Remove or mark visibly all obstacles, such as trash, large tree branches or stones, from the working area before operation. Check the area for possible hidden ditches, manholes, or other dangerous areas. Remember that tall grass can hide obstacles.
- Set and lock all four wheels of the mower to same height.
- Ensure that bystanders are at a safe distance when operating the equipment. Do not let anyone to enter danger area of the boom or to stay directly in front of the loader. Also make sure that it is safe to reverse with the loader. Never assume that bystanders will remain where you last saw them; especially children are often attracted to the moving equipment.
- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The attachment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Refer to Chapter 7 for maintenance instructions.
- Operate the attachment and the controls of the loader only when seated on the driver's seat. Ensure that the loader and the attachment are used in a safe manner and as intended. Do not allow children to operate the equipment.
- Never operate the loader or attachments while under the influence of alcohol, drugs, medication that may impair judgement or cause drowsiness, or if not otherwise medically fit to operate the equipment.
- Remember correct working methods and avoid leaving the driver's seat unnecessarily.

NOTICE

Regularly clean the loader oil cooler core; refer to the operator's manual of the loader for more instructions. Overheating oil decreases the power and affects the service life of the hydraulic components of both the attachment and the loader.

6.2 Using the mower

To start the lawn mower, turn the auxiliary hydraulics lever of the loader to its locked position. The mower can be stopped by releasing the lever. If the hydraulic hoses are connected incorrectly when using the conventional quick couplings, the lawn mower will not start running. Change the position of the hoses if necessary.

NOTICE

It is important to set the floating system as instructed on the following page. Otherwise the blades may not turn, or the mower will not follow the ground properly.

- Adjust engine RPM so that the loader's auxiliary hydraulic flow is within the specified range for the mower. Too little flow will not rotate the blades fast enough for good mowing results. However, too high setting will result in excess vibration and noise, and the mower may get damaged. Generally, only the 1-pump setting should be used (on applicable loaders).
- Adjust drive speed according to cutting height, grass type, and other conditions to ensure clean cut. the blades to rotate fast enough. Monitor the loading and rotating speed of the mower by listening to changes in its noise.
- Avoid cutting wet grass. Wet or dense grass can build up under the deck, gradually decreasing mower performance.
- Keep the blades in good condition, and keep the mower clean. Sharp blades will increase efficiency and improve the quality of cut.
- Avoid cutting too much at one pass. Generally, no more than one third of the length of the grass should be cut at once. Tall grass may require incremental cutting with gradually lower cutting height.
- Mowing in reverse direction is not recommended as the loader wheels compact the grass just before cutting while reversing.



The mower can throw stones, dirt, or other material at high velocity, also in mulching or collecting configurations. Keep a safe distance to other persons and stop the mower when there are stones or other objects on the ground. Never direct the side discharge chute (if fitted) in the direction of people, animals, buildings, vehicles, or fragile objects.



Always look in the direction the loader is moving. Before reversing, always ensure that bystanders, especially children, are at a safe distance to avoid back-over accidents.



Operating a powerful mower requires the full attention of the operator. Do not perform distractive actions while mowing, such as using mobile devices.

NOTICE

Stop the mower immediately if it makes abnormal noise, vibrates, or if the blades are blocked. Find out the cause before continuing use.

6.3 Working on uneven ground

Extra caution is needed when using the equipment on inclined terrains and slopes. Drive slowly especially on inclined, uneven, or slippery surfaces, and avoid sudden changes in speed or direction. Operate the controls of the loader with careful and smooth movements. Watch out for ditches, holes on the ground, and other obstacles, as hitting an obstacle may cause the loader to tip over.

Maximum lifting capacity can not be achieved on inclined terrain. On horizontally tilted terrain the load must not be lifted high. The loader frame articulation should be kept straight when lifting heavy loads; turning the load during lifting operation will affect the stability of the loader and may lead to overturning of the machine.

Mow slopes by driving straight up and down, not across. Avoid starting or stopping on a slope. Exercise extreme caution when changing direction on slopes.

If the tyres lose traction, stop the mower blades and proceed slowly straight down the slope. Do not mow near drop-offs, ditches or embankments. The loader could suddenly turn over if a wheel is over the edge of a cliff, ditch, or if an edge caves in.

6.4 Floating system

The mower is equipped with a mechanical floating system, which enables efficient mowing on uneven terrains. Floating refers to a linkage that allows the mower to follow the contours of the lawn without moving the loader boom.

Floating is adjusted by using the controls of the loader boom from the driver's seat, when ready to begin mowing.

To set the mower linkage to correct initial position:

- Position the loader boom and its quick coupling plate to about in the middle of the extreme positions of the floating linkage.
- The middle position in the adjacent figure indicates the correct initial setting.

The operation of floating is greatly affected by its setting. If the floating has been set too high, the mower is easily left hanging in the air. When it is too low, the loader's front wheels are easily raised off the ground.



NOTICE

When setting the mower floating linkage to correct operating position, avoid pressing the mower hard against the ground with the loader boom to avoid damaging the mower. Operate both the lift and tilt movements of the loader boom carefully.

If your loader is equipped with the optional boom floating system, it is advisable to switch it on, as this allows greater movement range than the lawn mower floating alone on uneven terrains. The loader boom floating should be switched on when the lawn mower floating is at its lowest position. See loader manual for further instructions about the boom floating system.

6.5 Handling of the clippings

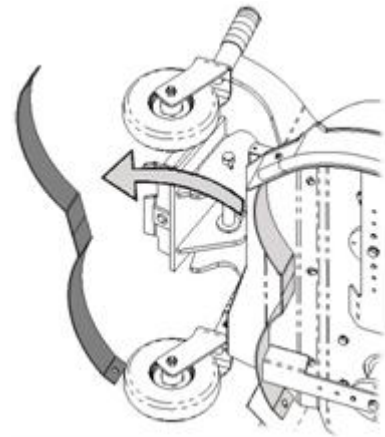
Mulching (standard configuration):

In standard configuration, the Lawn Mower 1000 will mulch the grass, leaving clippings evenly under the mower. Mulching allows to leave the shredded clippings on the ground when the lawn is mown often enough. The mulching mode works the best with dry grass and when not too much grass is cut at once.

Rear discharge mode:

If necessary, the lawn mower can be set to a rear ejection mode by removing a rear discharge cover. Rear discharge requires less power than mulching mode, so it is often practical in large areas, as it makes it possible to mow slightly longer grass and at higher travel speed.

The rear discharge operation is set by removing the rear cover plate shown in figure.



6.6 Adjusting the cutting height

The cutting height can be adjusted by using the plastic sleeves on the vertical axles of the bearing wheels. The mower does not need to be disconnected from the loader while adjusting the height. To adjust the height:

1. Lift the mower slightly off the ground with the loader and shut down the engine.
2. Remove the cotter pin from the vertical axle of the bearing wheel. Pull the wheel off its holder.
3. Move some plastic sleeves from above the bracket to below it or vice versa, depending on whether you wish to raise or lower the mower.
4. Push the axle back in through the bracket and reinstall the cotter pin.
5. Repeat steps 2–4 for each wheel. Ensure that all wheels are adjusted to the same height.



NOTICE

There are two thicknesses of plastic sleeves (10 & 20 mm) for more accurate adjustment of the cutting height. Adjust all four wheels to the same height.

6.7 Removing a blockage

It is unlikely that the cutting deck gets blocked in normal operation. However, when when mowing wet grass or driving at too high speed, the mower may get blocked. If blocking occurs, check the cutting deck and clean if necessary.



Always stop the loader by following the safe stopping procedure before attempting to remove blockages. Never reach to the cutting deck if the blades may start rotating.

7. Maintenance and service

The attachment has been designed to be as maintenance free as possible. Continuous maintenance includes regular cleaning and lubrication, and monitoring the condition of the attachment. Because of the crushing hazard caused by lowering machine parts, all maintenance work must be done when moving parts have been lowered down completely and the attachment is lowered flat against the ground.



Make sure the attachment is properly supported during all maintenance work. Never go under a lifted attachment. The loader boom may lower unexpectedly during maintenance causing serious injuries even when the loader engine is not running. All maintenance and service must be performed when the attachment has been lowered down to a safe position.

7.1 Checking hydraulic components

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Wash any part of body that has been in contact with hydraulic oil carefully with water and soap. Hydraulic fluid is also harmful to the environment and any leak to the environment must be prevented. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Operate the attachment only with type of hydraulic oil that is accepted for use in Avant loaders.



Never handle hydraulic components when the hydraulic system is pressurised, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. If there are signs of leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected. Do not use hands to search for leaks. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. Check the routing of the hoses; adjust the hose clamps to avoid abrasion to the hoses. The hoses have limited service life. Depending on operating conditions all hoses must be inspected thoroughly no later than after 3 to 5 years of use, and if required, they must be replaced with new ones.

Finding any fault means that the hydraulic hose or component must be replaced and the equipment must not be used until it is repaired. Spare parts are available from your nearest AVANT retailer or authorised service point. Leave the repair work to professional service technicians, if you don't have adequate knowledge and experience about hydraulic assemblies and how to perform the repairing safely.

7.2 Cleaning and lubricating the mower

Drying grass may build up to decrease the performance of the mower and should be removed after each use. The mower deck should only be cleaned manually to avoid damaging the bearings. Stop the loader and disconnect hydraulic couplings, then remove grass by using a stick or brush. If using a pressure washer, care must be taken to avoid spraying on the bearings and hydraulic components.

NOTICE

The mower can be cleaned with a pressure washer only if low pressure is used and spraying at the bearings and the hydraulic components is avoided. The bearings may get damaged if sprayed with high pressure.

On the Lawn Mower 1000 there are lubrication points only on the vertical wheel support brackets. Add a small amount of grease to these 1–3 times a year. Clean the end of the nipple before greasing and add only a small amount of grease at a time. All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples. The bearings of the mower blades are maintenance free and can't be greased. The floating linkage does not normally require lubrication, but if seen necessary, apply oil on the linkage joints.

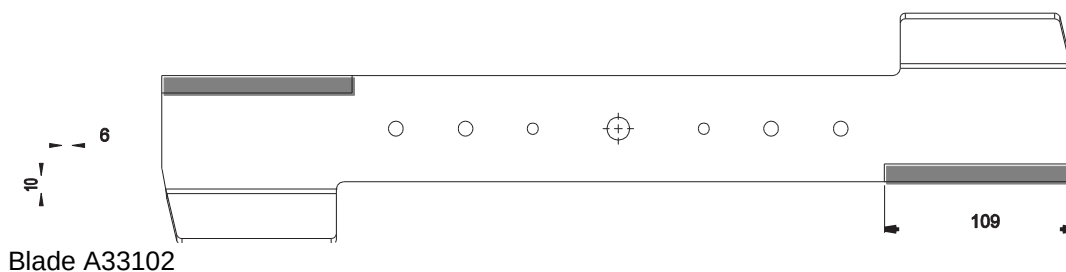
7.3 Sharpening and replacing the cutting blades

The blades wear down during normal use and the condition of the blades should be checked after every few uses. If the mowing deck is clean, signs that point to dull or damaged blades include the following:

- Decreased quality of cut, resulting in uneven cutting, torn grass, or yellowish grass after mowing
- Decreased performance / increased power requirement
- Increased vibration or noise

Evenly worn blades can be sharpened, but if there are deep notches, cracks, rock impact marks or similar signs of damage, the blades must be replaced with new ones. To retain balance, sharpen all blade surfaces in the same manner.

Do not sharpen the blades too sharp, as knife like sharpened blades will wear down very quickly. Note that the blade temperature should remain as steady as possible during sharpening; excess heating during sharpening will weaken the strength and wear resistance of the steel. Check the balance of the blades after sharpening. Only use blades identical to the original ones; use only original spare parts. Note the length of the blade sharpening area as indicated in the figure below.



NOTICE

If the blades have large notches or cracks, or if they have worn down too much or unequally, all blades must be replaced. Always sharpen or replace all blades at the same time.



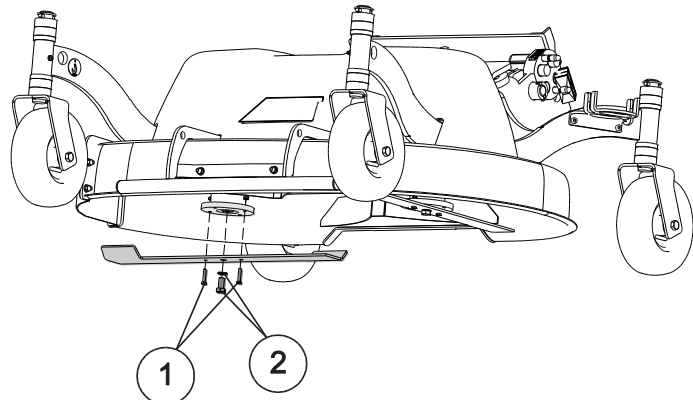
Always use good protective gloves when handling the blades. Ensure that the mower is properly supported during inspection and when replacing the blades.

7.4 Fastening of the blades

The blades are connected with a central bolt and two shear bolts. To protect the blades and the bearings of the mower, the shear bolts break or bend if the blades hit a rock or other heavy obstacle.

The shear bolts and the timing of the blades must be checked after the blades collide heavily with an obstacle. The shear bolts must be replaced with new ones when they show signs of deformation.

1. Two shear bolts, regular grade 8.8 M6x25 screws, secured with nyloc locking nuts.
2. Centre bolt with locking washer. When fitting the blades back to the mower, a lock washer must be fitted under each bolt.

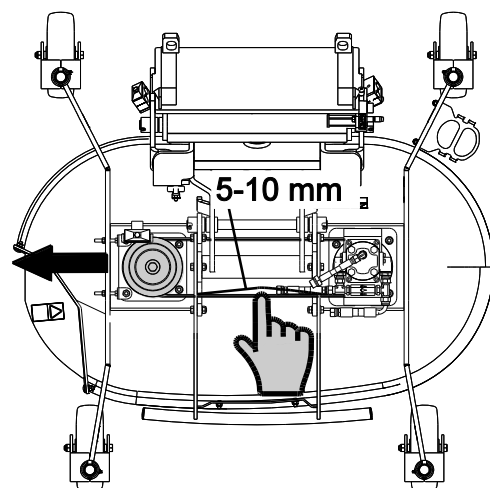


7.5 Checking and tightening the drive belt

The tightness and the condition of the drive belt must be checked at least annually at the beginning of each mowing season, and depending on use, a few times during the season. A belt that is not at its correct tightness may not necessarily be apparent to the user, but sudden slipping or breaking of the belt may lead to a hazardous situation or damage to the mower or the blades.

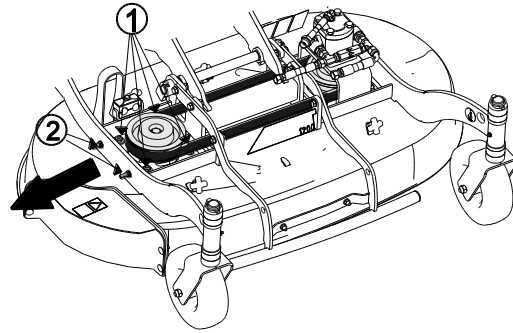
To check the belt, remove the protective cover from the top of the mower.

- Push the belt horizontally at its longest section between the pulleys with a force of about 50-80 N. This equals to the force of about 5-8 kg weight.
- The belt should move about 5-10 mm when the load is applied horizontally. Check that the belt is tightened correctly and not too much as well.
- Check the condition of the belt visually; if the belt is clearly worn or material has come off, the belt must be replaced.



If necessary, the belt is adjusted by moving the belt pulley.

1. Loosen the four bolts holding the pulley base at the other end of the belt.
2. Tighten bolts 2, until the belt is correctly tightened.
3. Tighten the four retaining bolts. recheck belt tightness.
4. Refit the protective cover.



NOTICE

Check the synchronization of the blades after tightening the belt and replace all protective covers and guards before using the mower.

7.6 Checking the timing of the blades

The blades overlap partially as they rotate. It is important that the blades are correctly synchronized; otherwise the blades will collide with each other. The timing has been correctly set at the factory, but it may have to be checked and adjusted, if the blades hit a large rock or other obstacle.

NOTICE

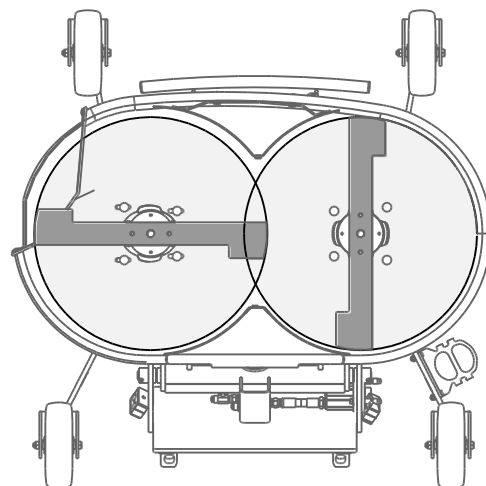
If the mower makes a constant clacking noise, the mower must be stopped and the timing of the blades must be checked immediately.

NOTICE

Keeping the belt in correct tightness will reduce the possibility of the misalignment of the blades.

Checking the blade synchronization:

1. Lift the mower slightly off the ground and turn the loader engine off. Follow safe stopping procedure.
2. Rotate the blades manually. Set one of the blades to its vertical position. The other blade should be perpendicular to it, as shown in adjacent figure.
3. If the blades are not perpendicular to each other, the synchronization must be adjusted by loosening the toothed belt as described in previous chapter.



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
 EG-försäkran om överensstämmelse
 EU samsvarserklæring
 EF-overensstemmelseserklæring
 EC Declaration of Conformity
 EG-Konformitätserklärung
 Déclaration de conformité CE



Valmistaja / Tillverkare / Produsent / Producent / Manufacturer / Hersteller AVANT TECNO OY
 Fabricant:

Osoite / Adress / Adresse / Address / Adresse: Ylötie 1
 33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu /

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats /

Vi erklærer herved at produktet som er oppgitt under er i samsvar med forskriftene i Maskindirektivet (direktiv 2006/42/EC med endringer). Følgende harmoniserte standarder har blitt anvendt /

Vi erklærer herved, at nedenstående produkter er i overensstemmelse med bestemmelserne i maskindirektivet (direktiv 2006/42/EF indeholdende ændringer). Følgende harmoniserede standarder er anvendt /

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied /

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

Nous déclarons par la présente que les produits mentionnés ci-dessous sont conformes aux exigences en matière de sécurité et de santé de la directive relative aux machines (directive 2006/42/CE, avec ses modifications) Les normes harmonisées suivantes ont été appliquées

SFS-EN ISO 12100, SFS-EN ISO 4413

Mallit / Modeller / Modeller / Modeller / Models / Modellen:

Avant	
Hydraulitoiminen ruohonleikkuri; Avant-kuormaajan työlaite	
Hydraulisk gräsklippare; arbetsredskap för Avant lastare	
Hydraulisk lenklipper; redskap for Avant minilastere	A21076
Hydraulisk græsklipper redskab til Avant Minilæssere	A21109
Hydraulic lawn mower; attachment for Avant loaders	
Hydraulische Rasenmäher; Anbaugerät für Avant Radlader	
Tondeuse à commande hydraulique destinée à être utilisée avec les chargeuses Avant	



4.8.2015 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör / Administrerende direktør /
 Managing Director / Geschäftsführer / Directeur général



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